

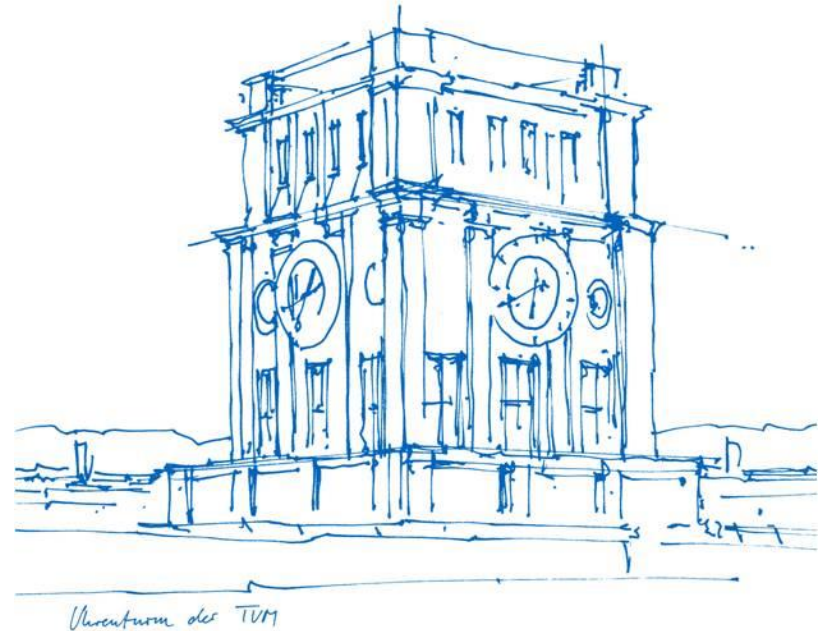
Advanced Practical Course: Development of AI and Data-driven Applications at the Example of ERPsim

Project Kick-Off – Summer Term 2026

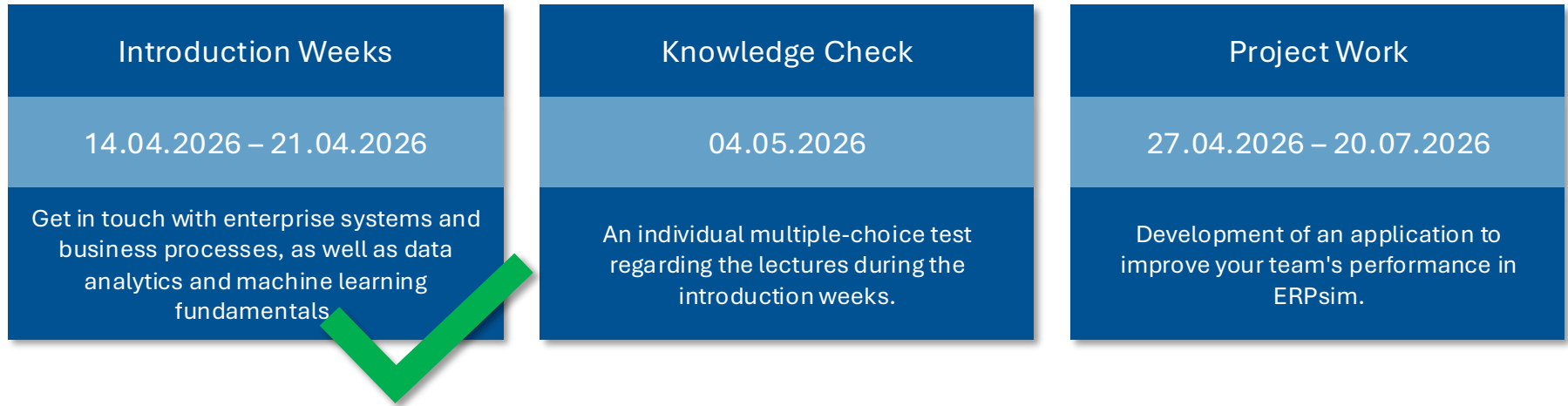
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Information Systems and Business Process Management
TUM School of Computation, Information and Technology
Technical University of Munich

April 28th, 2026



Building Blocks of this Practical Course



Your task for today



Form your final project groups where you will work with for the remaining semester. Please send us the names of the group members and your company letter by email until end of day. Each team consists of four students.

Part 2: Knowledge Check

Knowledge Check

04.05.2026

An individual multiple-choice test regarding the lectures during the introduction weeks.

Important Date for the Knowledge Check

Monday

04.05.2026

Knowledge Check



Be on time!

Otherwise, you will be excluded from the test.

Further Information

- Individual multiple-choice test (mainly) in Moodle
- Questions about the lectures during the introduction weeks
 - Focus on high-level concepts
 - No questions about navigation in tools
- 15 minutes for 15 questions
- You need seven correct questions to get a 4.0
- Makes up 10 points of your final grade

Part 3: Project Work – Overview Appointments

Project Work

27.04.2026 – 20.07.2026

Development of an application to improve your team's performance in ERPsim.

Important Dates for the project work

Project Introduction Sessions

Monday	27.04.2026	09:00 – 12:00	FMI 01.13.007
Tuesday	28.04.2026	09:00 – 12:00	FMI 03.09.012

Concept Presentation

Monday	04.05.2026	15:00– 17:00	SAP Labs
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Agile Workshop

Thursday	07.05.2026	09:00 – 17:00	tba.
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ERPsim playing Sessions

Monday (1)	11.05.2026	09:00 – 12:00	FMI 01.13.010
Thursday (1)	28.05.2026	09:00 – 12:00	FMI 01.13.010
Tuesday (2)	09.06.2026	09:00 – 12:00	FMI 01.09.014
Wednesday (2)	24.06.2026	09:00 – 12:00	FMI 01.13.010
Tuesday (1)	07.07.2026	09:00 – 12:00	tba.

Project Submission Deadline

Sunday	19.07.2026	23:59	
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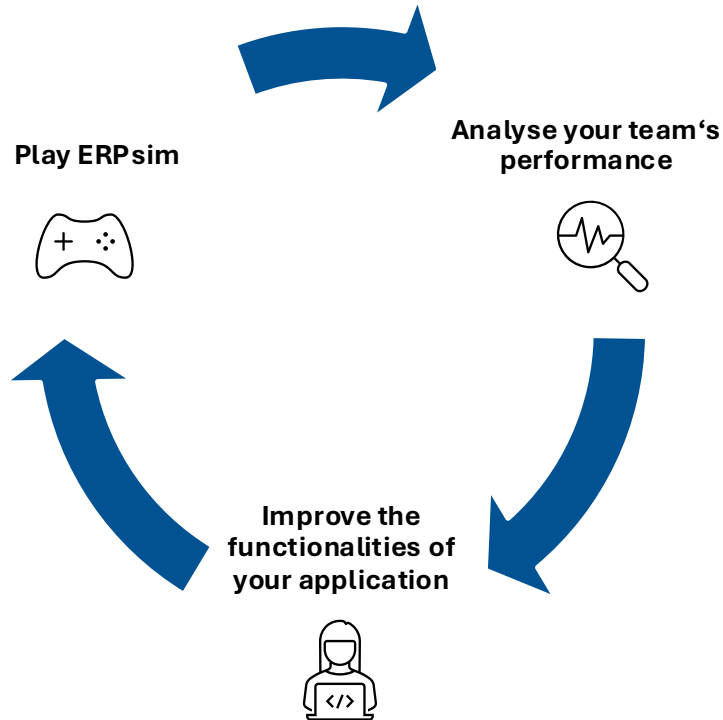
Final Presentation

Monday	20.07.2026	09:00 – 12:00	SAP Labs
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Part 3: Project Work – Overview Concept



Develop an analytical application that support the decision-making process to improve your team's performance in ERPsim.

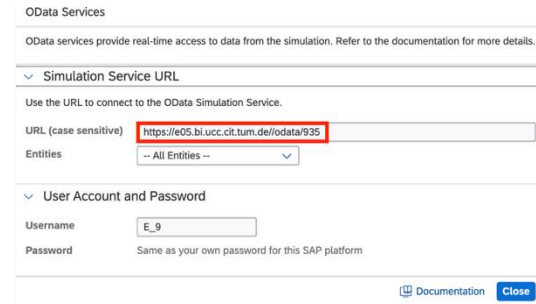
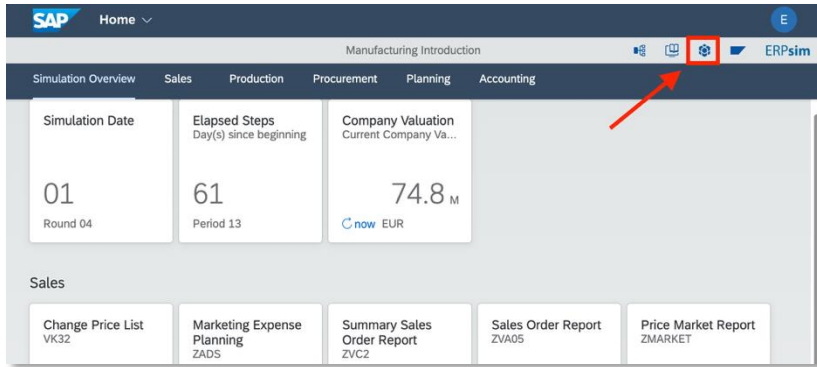


One cycle takes two weeks

7x

Playing seven rounds during the semester

Part 3: Project Work – Game Data Source



```
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```

ERPsim provides you important game data via an OData service you can integrate in many tools (among others Excel for ad-hoc analytics)

Part 3: Project Work – Disruptor Overview

Cold Spell

- **Increase** carbon emissions for **2 raw materials**
- **Decrease** carbon emissions for **overstocking**
- **Decrease** carbon emissions for **purchased energy**

Heat Wave

- **Increase** carbon emissions for **2 raw materials**
- **Increase** carbon emissions for **overstocking**
- **Increase** carbon emissions for **purchased energy**

Supplier Disruption

- **Increase** carbon emissions per **delivery**
- **Increase** carbon emissions per **unit purchased**

Carbon Decree

- **Increase carbon tax**
- **Decrease** carbon emissions for **purchased energy**.

Energy Recovery

- **Increase** carbon emissions for **setup time**
- **Decrease** carbon emissions per **unit produced**
- **Decrease** carbon emissions for **purchased energy**

Hub Relocation

- **Increase/decrease** carbon emissions for **internal transfers**.
- **Increase** carbon emissions for **sales from the central warehouse**

Flash Flood

- **All products in the main warehouse gets lost.**

Part 3: Project Work – Data Sources for Disruptions



Energy Market Information
Water Level Information
Roadwork News

**REST
Endpoint**

Weather Information
Supplier Stock Market



S3 Bucket

EU Legal Documents

Part 3: Project Work – Project Expectations

Project Work

27.04.2026 – 20.07.2026

Development of an application to improve your team's performance in ERPsim.

Expectations about the project

- Apply one of the **data architecture concepts** (e.g., Data Lake, Lakehouse, Data Fabric, or comparable concept)
- Structured use of LLMs, following a clearly defined **agent pattern** (if applicable)
- **High-quality and maintainable code**
- Clear and **explicit linkage to the game**, especially to in-game decision-making
- **Well-argued justification** of the proposed idea with respect to both the game and the system architecture
- Implementation of backend and frontend **using CAP**

Part 3: Project Work – Concept Presentation

Present the concept of your application.

04.05.2026

Naming: SoSe26_[COMPANY-
LETTER]_concept_presentation

15 min (presentation)
+ 15 min (discussion)

(non-compliance will influence your grade)

You will be graded as a team
(no individual assessment)

Submit in pptx and PDF Format!

Upload your presentation until
03.05.2026 EOD via Moodle

The presentation needs to be held in English.

The following aspects must be included in the progress presentation:

- Team introduction (photos would be nice 😊)
- First application **architecture concept**
- Overview of **planned tools and technologies**
- Clear definition of the **ERPsim decision areas** to be supported (e.g., pricing, planning, marketing, or investment planning – choose wisely)
- Chosen project management approach based on Scrum, including defined **team roles**

Discussion (also graded): Questions, Notions, Propositions

Participation in the
discussion of other
teams is also graded!

Part 3: Project Work – Bi-weekly Status Presentation

Present the status of your application.

Every playing session

Naming: SoSe26_[COMPANY-
LETTER]_biweekly_presentation_[number]

5 min (presentation)
+ 10 min (discussion)

(non-compliance will influence your grade)

You will be graded as a team
(no individual assessment)

Submit in pptx and PDF Format!

Upload your presentation **before** the
session starts via Moodle.

The presentation needs to be held in English.

The following aspects must be included in the bi-weekly status presentation:

- **Exactly one slide** summarizing all relevant key information (poster-style overview)
- **Clear reflection on progress**, including what was done, what was expected, and whether those expectations were met
- **Presented at the end of each session**
- **All groups must submit every time**. The teaching team selects the groups to present per session

Discussion (also graded): Questions, Notions, Propositions

Participation in the
discussion of other
teams is also graded!

Part 3: Project Work – Final Presentation

Present the final state of your application.

20.07.2026

Naming:

SoSe26_[COMPANY-LETTER]_final_presentation

30 min (presentation)
+ 15 min (discussion)

(non-compliance will influence your grade)

You will be graded as a team
(no individual assessment)

Submit in pptx and PDF Format!

Upload your presentation until
19.07.2026 EOD via Moodle.

The presentation needs to be held in English.

The following aspects must be included in the presentation:

- Project Solution in the form of an **elevator pitch**
→ explaining why other teams should adopt or “buy” the solution
- Team introduction (photos would be nice 😊)
- Presentation of the final architecture, including key changes and the rationale behind them
- **Live demo** of the implemented solution
- Clear overview of the **five sprints (agile roadmap)**, including delivered outcomes and assigned roles per sprint
- Explanation of **how disruptions were handled** during the simulation
- **Estimated impact** on ERPsim performance, made measurable where possible (required for top grades)
- Main challenges during the project
- Reflection on lessons learned throughout the project

Participation in the discussion of other teams is also graded!

Discussion (also graded): Questions, Notions, Propositions

Part 3: Project Work – Documentation

Naming:

SoSe26_[COMPANY-LETTER]_documentation

The documentation must be written in English.

Limit to 15 pages

Please refer to Moodle for the template.

Submit in PDF format!

Upload your documentation until
19.07.2026 EOD via Moodle.

- **The following aspects must be included in the documentation:**

- **Project motivation and goal**
 - Theoretical background (for used algorithms and concepts; if applicable)
 - **Solution concept** and software architecture
 - Semantic and technical data models
- **Implementation details** (technical description of implemented features)
 - ER diagram, UML diagram, BPMN diagram etc.
 - Only explain specific code snippets for key aspects if necessary
- **Agile roadmap** of your project (including a comparison to original roadmap and responsibilities)
- Challenges during the project and lessons learned
- References (if applicable)

Submission of prerequisites and code collection of your application via GitLab from LRZ!

Agile Workshop Session on 07.05.26

Workshop covering agile project management by experts working at Deloitte:

In addition to agile project management, you will learn how to adapt the learned aspects to your projects including an agile roadmap.

(It is expected that you apply the learned concepts in your project.)

Good Luck!

Please don't hesitate to contact us in case of any question: sap-analytics-i17@tum.de